Function families

Topics: polynomial, exponential, logarithmic, and trigonometric functions.

Goal: learn the main types of functions, recognise their graphs, and compare how quickly they grow.

Today's question: **What can the shape of a function tell us before we calculate?**

Session
4 of 15

Date
Thu 10 Sep 2026

COURSE OUTLINE

Where session 4 sits

1 · Course overview and algebra refresher	2 · Sets and set operations
3 · Functions	**4 · Function families**
5 · Theorem proving	6 · Mathematical induction
7 · Combinatorics and the binomial theorem	8 · Midterm
9 · GCD and divisibility	10 · Modular arithmetic
11 · Vectors	12 · Polynomials
13 · Complex numbers	14 · Course recap and exam preparation
15 · Final exam	

TODAY Polynomial, exponential/logarithmic, and trigonometric families.

Functions, in one page

ANATOMY	
$f : A \to B$	Domain, codomain, one output each
graph	In $A \times B$, each a once
fails	No output, or two
$f(A)$	Image: what is hit
codomain	Promised; the image is proved
$f^{-1}(S)$	Preimage: set to set, always
$g \circ f$	$g(f(a))$, f first. Sets match
order	$g \circ f \neq f \circ g$ usually

BEHAVIOUR	
Injective	$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$: never twice
Surjective	Every b is hit
Bijective	Both: each b once
to prove	Assume equal images, or construct a
to disprove	One witness suffices
f^{-1}	Iff f bijects. Not $1/f$
restrict	Shrink until bijective; gives $\arcsin$
$\cos\theta, \sin\theta$	On $x^2 + y^2 = 1$; period 2π

WARM-UP Eventually larger: x^2 or 2^x? Why isn't $x = 1$ enough?

One transformation language works for every family

DEFINITION Start with the graph $y = f(x)$. For

$$g(x) = a\,f\big(b(x-h)\big) + k \qquad (b > 0),$$

read from the **inside out**: change x first, then change y.

EXAMPLE If $f(x) = x^2$, then $f(x-2) = (x-2)^2$: the vertex moves from $(0,0)$ to $(2,0)$.

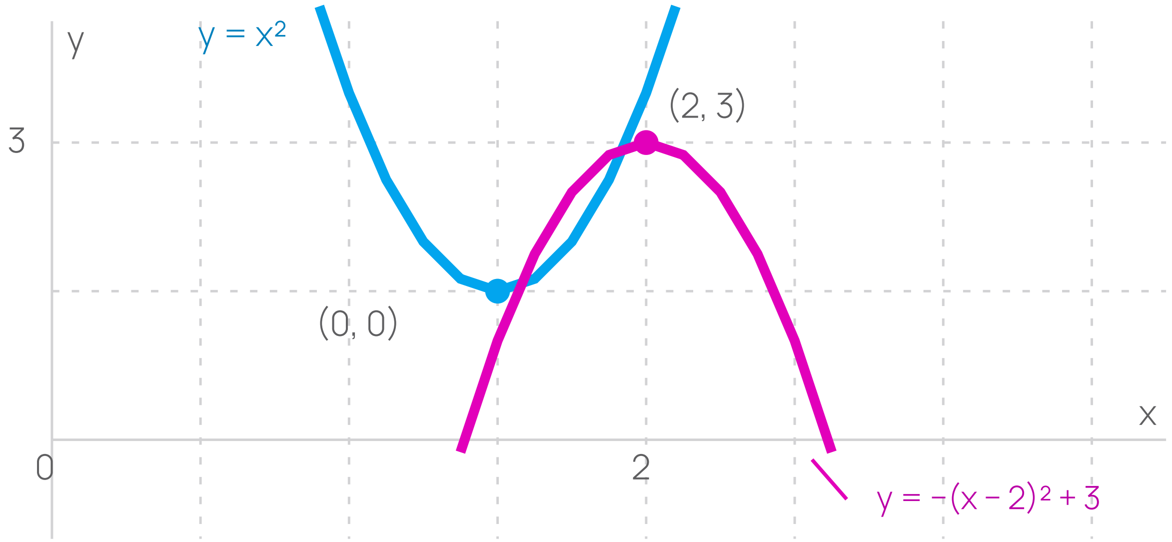
CHANGE	EFFECT ON THE GRAPH
h	move h units horizontally (right if $h > 0$)
k	move k units vertically (up if $k > 0$)
$\lvert a\rvert$	make the graph $\lvert a\rvert$ times taller; if $a < 0$, turn it upside down
b	make the graph $1/b$ times as wide

Track one familiar graph as it moves

EXAMPLE Start with $f(x) = x^2$. The graph of

$$g(x) = -(x-2)^2 + 3$$

moves right 2, turns upside down, and moves up 3. The vertex moves from $(0, 0)$ to $(2, 3)$.



FUNCTION FAMILY I

Polynomials

The formula tells us about the graph. Examples: $3,\ -2x+5,\ x^2-4x+3,\ 2x^5-x$.

In a simplified polynomial, the variable is not in a denominator, under a root, or in an exponent.

Question: Which part of a polynomial tells us what its graph does far left and far right?

1

Degree is the largest exponent with a coefficient that is not zero

DEFINITION A polynomial has the form

$$p(x) = a_n x^n + \cdots + a_1 x + a_0,$$

where $n \geq 0$ and $a_n \neq 0$. Its **degree** is n, and a_n is its **leading coefficient**. For this session, the coefficients a_i are real numbers.

WORKED

POLYNOMIAL	DEGREE	LEADING COEFFICIENT
$-4x^3 + 7x - 1$	3	-4
$5x^2 - 2x^5$	5	-2
6	0	6

METHOD Write the largest power first before you find the degree.

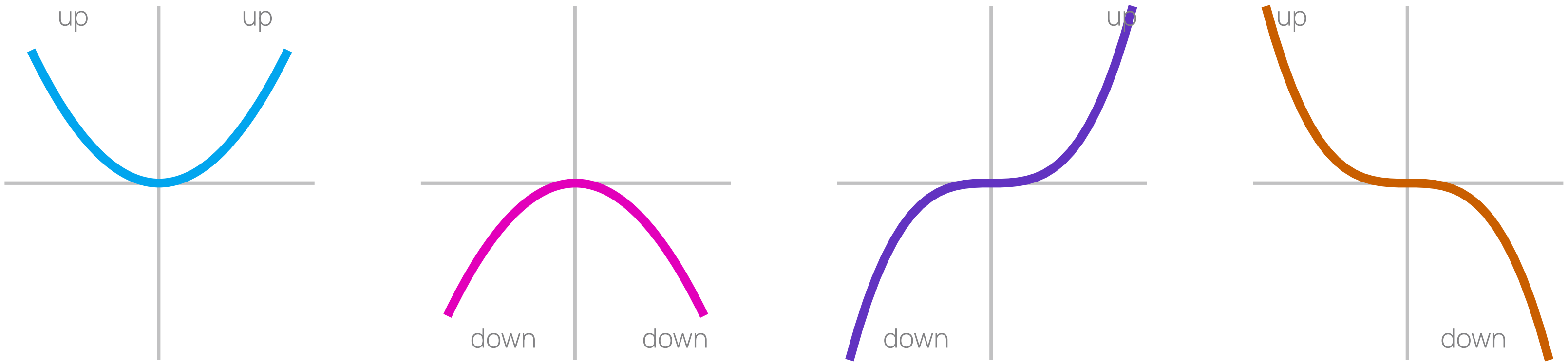
The highest-power term tells us what happens at the ends

DEGREE n	LEADING COEFFICIENT a_n	LEFT END	RIGHT END
even	positive	up	up
even	negative	down	down
odd	positive	down	up
odd	negative	up	down

READING IT When x is very large positive or very large negative, $p(x)$ behaves like $a_n x^n$. Here **up** means the value becomes very large positive; **down** means it becomes very large negative.

EXAMPLE $p(x) = -2x^5 + 7x^2 - 1$ has odd degree and a negative leading coefficient, so it goes **up on the left** and **down on the right**.

Four leading terms give the four possible end patterns



even, positive: x^2 | even, negative: $-x^2$ | odd, positive: x^3 | odd, negative: $-x^3$

USE THIS WITH THE TABLE Ignore the middle of a polynomial graph here. The degree's parity and the leading coefficient determine only what happens far to the left and right.

Roots are where the graph meets the x-axis

DEFINITION A number r is a **root** (or zero) of p when $p(r) = 0$.

FACTOR THEOREM

$$p(r) = 0 \iff (x - r) \text{ is a factor of } p(x).$$

EXAMPLE

$$p(x) = x^3 - 4x = x(x^2 - 4) = x(x-2)(x+2),$$

so the roots are -2, 0, and 2.

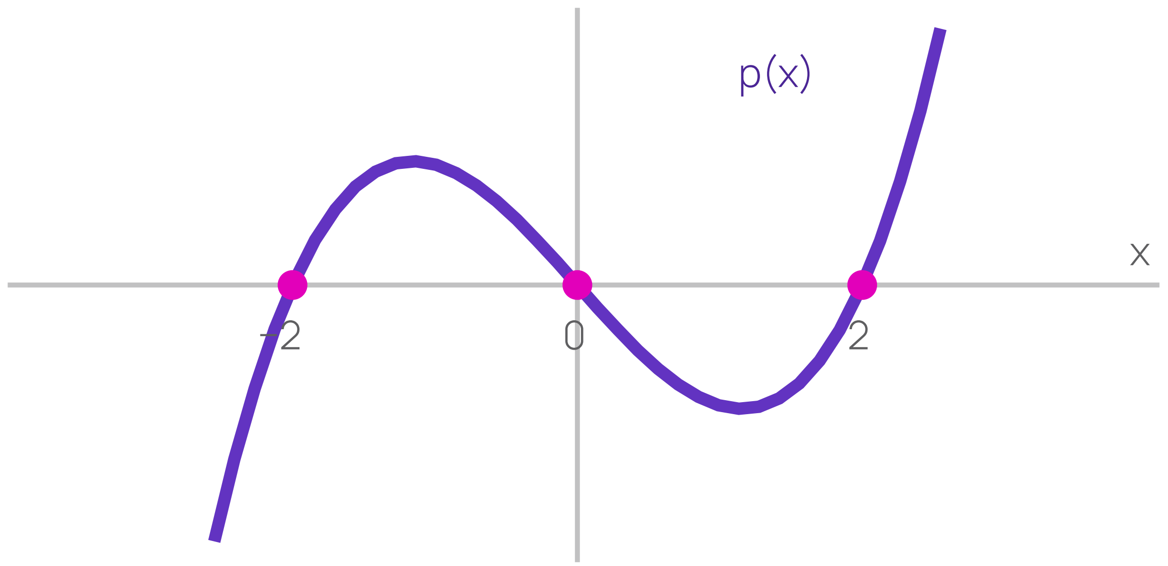
AT MOST A nonzero polynomial of degree n has **at most n different real roots**.

For this cubic, each root is an x-axis crossing

READ THE CROSSINGS

$$p(x) = x^3 - 4x = x(x-2)(x+2)$$

The graph crosses the x-axis at $x = -2, 0,$ and 2. Those x-coordinates are the roots.



ACTIVITY 1

Read a quadratic from its formula

ACTIVITY 1 GROUPS OF THREE 12 MIN

For $q(x) = -(x-2)^2 + 3$:

1. State the degree and leading coefficient.
2. Describe the transformation from $y = x^2$.
3. State the vertex and end behaviour.
4. Find the roots in exact form (do not use decimals).

Share: one group explains why the two roots are $2 \pm \sqrt{3}$, rather than $\pm\sqrt{3}$.

Work outwards from the completed square

METHOD

1. Expand mentally to get the degree; the number in front of the square is the leading coefficient.
2. Match q against $a\,f\big(b(x-h)\big)+k$ with $f(x)=x^2$: read h, then k, then a.
3. The vertex sits where the square is zero, so solve $x-h=0$.
4. For roots, isolate the square and take **both** square roots.

ANSWER Degree 2, leading coefficient -1; shift right 2, up 3, then reflect vertically; vertex $(2,3)$; both ends down.

$$-(x-2)^2+3=0 \iff (x-2)^2=3$$

so $x-2=\pm\sqrt{3}$ and $x=2\pm\sqrt{3}$.

THE TRAP Answering $\pm\sqrt{3}$ drops the horizontal shift. Undo the square first, then undo the shift.

FUNCTION FAMILY II

Exponentials and logarithms

Repeated multiplication and its inverse. If adding 1 to the input always multiplies the output by the same number, the pattern is exponential.

$$2^0 = 1, \quad 2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8$$

A logarithm answers the reverse question: $2^x = 8$ asks: "What exponent gives 8?"

2

An exponential function has the variable in the exponent

DEFINITION

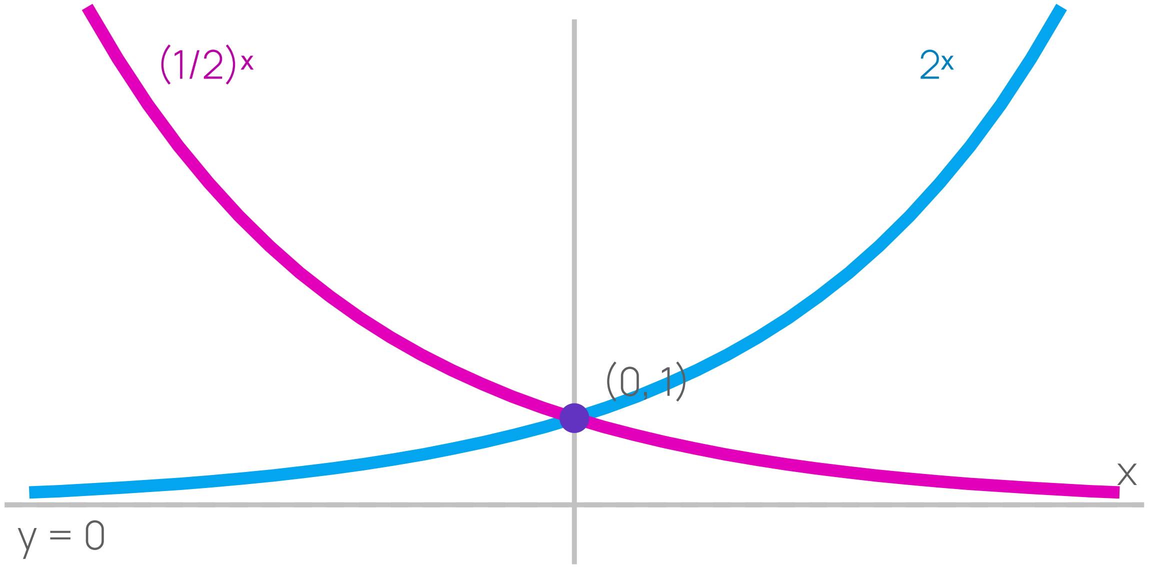
$$f(x) = b^x, \qquad b > 0,\ b \neq 1$$

Every exponential graph passes through $(0, 1)$ because $b^0 = 1$. The line $y = 0$ is a horizontal **asymptote**: the graph approaches it but never reaches it.

BASE	BEHAVIOUR	DOMAIN AND RANGE
$b > 1$	growth	domain $\mathbb{R}$, range $(0, \infty)$
$0 < b < 1$	decay	domain $\mathbb{R}$, range $(0, \infty)$

Growth and decay share a point, then move in opposite directions

COMPARE Both $y = 2^x$ and $y = (1/2)^x$ pass through $(0, 1)$. For every real x, both stay above the x-axis. The dashed line is the asymptote $y = 0$.



An exponential model uses the same multiplier at every step

MODEL

$$N(t) = N_0 r^t,$$

where $N_0 > 0$ is the starting amount and $r > 0$ is the multiplier per time unit.

MULTIPLIER r	INTERPRETATION
$r = 1.20$	increase by 20% per time unit
$r = 0.85$	decrease by 15% per time unit

EXAMPLE $N(t) = 100(1.2)^t$ gives $N(3) = 172.8$. The time unit must be named: hours, days, years, or something else.

Make the bases match, then set the exponents equal

RULE For $b > 0$ and $b \neq 1$,

$$b^u = b^v \implies u = v.$$

WORKED EXAMPLE

$$\begin{aligned} 3^{2x-1} &= 27 \\ 3^{2x-1} &= 3^3 \\ 2x - 1 &= 3 \\ x &= 2 \end{aligned}$$

Each line has the same solution as the line before it.

A logarithm is an exponent

DEFINITION For $b > 0$, $b \neq 1$, and $x > 0$,

$$\log_b x = y \iff b^y = x.$$

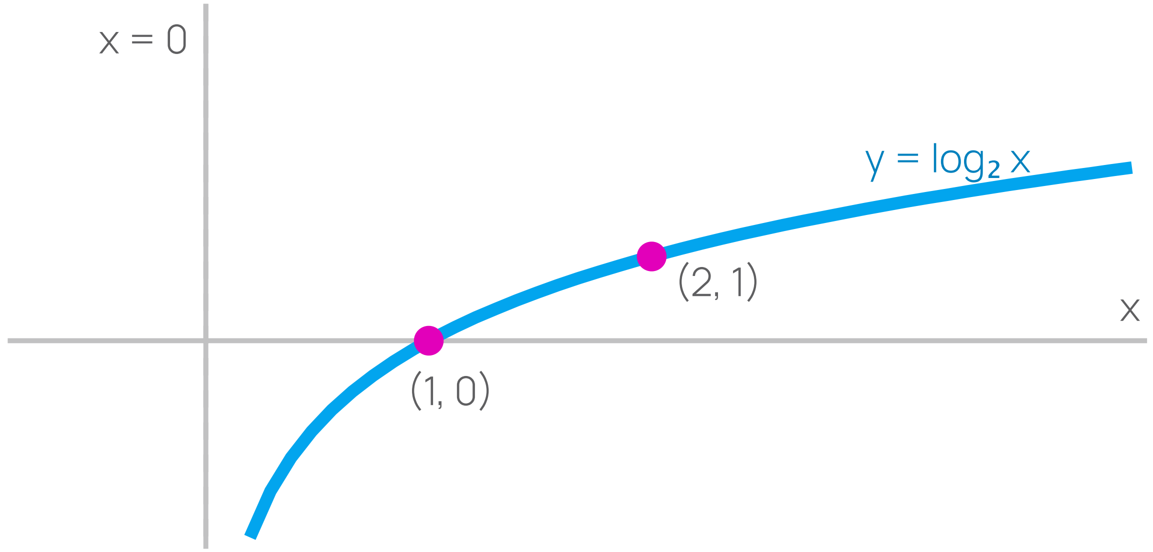
EXAMPLES

$$\log_2 8 = 3, \qquad \log_{10}(0.01) = -2, \qquad \ln(e^5) = 5.$$

DOMAIN $y = \log_b x$ has domain $(0, \infty)$ and range $\mathbb{R}$. There is no real value for $\log_b 0$ or $\log_b(-4)$.

The logarithm graph explains its domain

READ THE GRAPH $y = \log_2 x$ crosses at $(1, 0)$ and passes through $(2, 1)$. It exists only to the right of the y-axis: x must be positive.



Log laws change multiplication into addition

LAWS For $u > 0$ and $v > 0$:

$$\log_b(uv) = \log_b u + \log_b v$$

$$\log_b\left(\frac{u}{v}\right) = \log_b u - \log_b v$$

$$\log_b(u^c) = c\log_b u$$

THIS IS USUALLY FALSE

$$\log_b(u+v) \neq \log_b u + \log_b v \quad \text{in general.}$$

Counterexample: with $u = v = 1$, the two sides are $\log_2(1+1) = 1$ and $\log_2 1 + \log_2 1 = 0$.

ACTIVITY 2

First find the values of x that are allowed

ACTIVITY 2 PAIRS 14 MIN

Solve over the real numbers: $\log_2(x-1)+\log_2(x+1)=3.$

1. Find the values of x allowed by both logarithms.
2. Combine the logarithms using a log law.
3. Solve the resulting equation.
4. Check every possible solution in the original equation.

Share: explain why $x=-3$ is rejected even though it solves the squared equation.

Domain first, then algebra, then check

METHOD

1. Write the domain before touching the equation: every $\log_2$ input must be positive.
2. Combine the two logarithms with the product law, which needs both inputs positive.
3. Undo the logarithm using $\log_2 A = 3 \iff A = 2^3$.
4. Test every candidate against step 1, then substitute into the original equation.

ANSWER $x - 1 > 0$ and $x + 1 > 0$ give $x > 1$. Then $\log_2\big((x-1)(x+1)\big) = 3$, so $x^2 - 1 = 8$, $x^2 = 9$, $x = \pm 3$. Only $x = 3$ lies in the domain. Check: $\log_2 2 + \log_2 4 = 1 + 2 = 3$.

THE TRAP $x = -3$ solves $x^2 = 9$ but not the original equation — squaring is where extra solutions appear.

Change a log base; compare growth only for large n

BASE CHANGE When $x > 0$, $a > 0$, $b > 0$, and neither base is 1:

$$\log_b x = \frac{\log_a x}{\log_a b}.$$

So changing the log base only multiplies the answer by a constant.

GROWTH For fixed $d > 1$, $k > 0$, and $c > 1$, when the positive integer n becomes large,

$$\log_d n \; < \; n^k \; < \; c^n$$

This is called an eventual comparison: the inequality is true once n is large enough, but it may not be true for every $n \geq 1$.

ACTIVITY 3

Rank the functions when n is large

ACTIVITY 3 GROUPS OF THREE 12 MIN

Put these in order from slowest growth to fastest growth when n is large: $\log_2 n, \quad n, \quad n\log_2 n, \quad n^2, \quad 2^n.$

1. Why is $n\log_2 n$ larger than n once n is large?
2. Why does one numerical example not prove the full ranking?
3. Which expression is most likely to grow too quickly in an algorithm?

Share: one group gives the complete ranking and says why "when n is large" matters.

Compare neighbours, not all five at once

METHOD

1. Commit to a candidate order first; then you only have four adjacent pairs to justify.
2. For each pair, divide one by the other and ask what happens as n grows.
3. Name the n from which your claim holds — that is what "eventually" means.

ANSWER

$$\log_2 n \;<\; n \;<\; n\log_2 n \;<\; n^2 \;<\; 2^n$$

For $n > 2$, $\log_2 n > 1$, so $n\log_2 n > n$. And $n\log_2 n < n^2$ because $\log_2 n < n$. The exponential is the dangerous one for large inputs.

THE TRAP At $n = 2$, $n^2 = 4 = 2^n$. One value can tie or even reverse the order, so it proves nothing.

FUNCTION FAMILY III

Trigonometric functions

Graphs that repeat. From the unit circle, $-1 \le \sin x \le 1$ and $-1 \le \cos x \le 1$ for $x \in \mathbb{R}$;

and, in radians, $\sin(x + 2\pi) = \sin x$ and $\cos(x + 2\pi) = \cos x$.

Repeating graphs can model cycles: daylight, tides, sound, rotation, and seasonal demand.

3

Read a sine model from its parameters

DEFINITION

$$y = A\sin\big(B(x - C)\big) + D, \qquad A \neq 0,\ B > 0$$

If $A < 0$, the wave is turned upside down before it is moved by D.

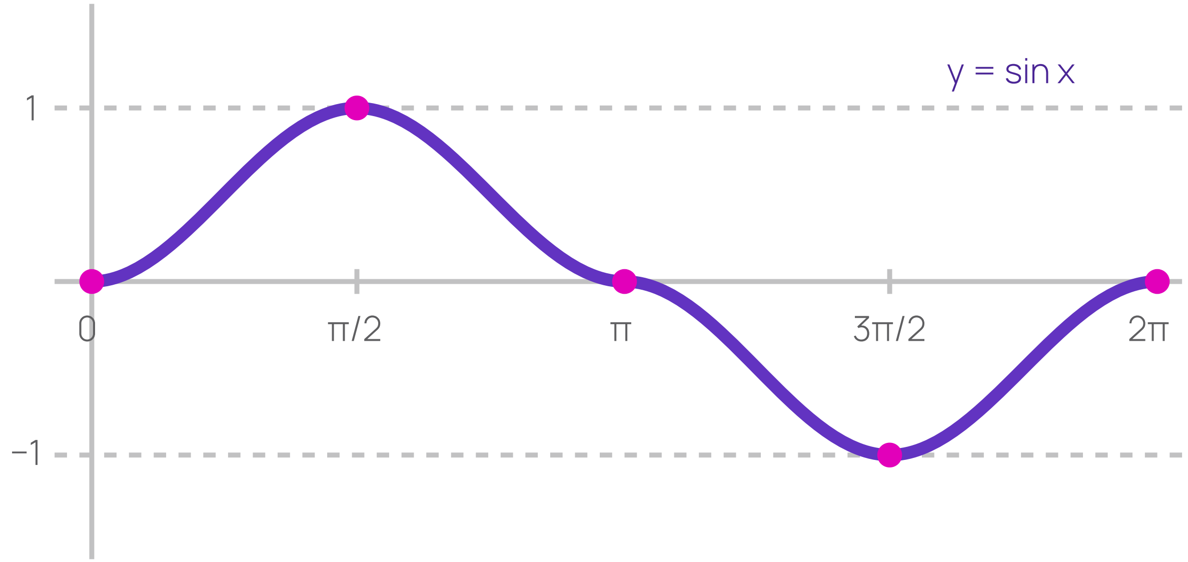
THE PARAMETERS

FEATURE	VALUE
midline	$y = D$
amplitude (distance from the midline to a peak)	$\lvert A\rvert$
period	$\frac{2\pi}{B}$
horizontal shift	move C units horizontally (right if $C > 0$)
range	$[D - \lvert A\rvert,\ D + \lvert A\rvert]$

One sine cycle gives five useful landmarks

QUARTER-PERIOD STEPS For $y = \sin x$, start at $(0, 0)$

. Moving one quarter period at a time gives a peak, a midline crossing, a trough, and the next crossing. The full cycle has length 2π.



Worked model: daily temperature

MODEL Let t be hours after midnight. Consider

$$T(t) = 4\sin\left(\frac{\pi}{12}(t-3)\right) + 28.$$

The maximum occurs at $t = 9$ hours; the minimum occurs at $t = 21$ hours.

FEATURE	INTERPRETATION
amplitude 4	temperature varies 4° from its average
midline 28	average temperature is 28°
period 24	one complete cycle takes 24 hours
range $[24, 32]$	lowest and highest temperatures predicted by the model

ACTIVITY 4

Interpret before calculating

ACTIVITY 4 PAIRS 12 MIN

For $H(t) = 1.5\cos\left(\frac{\pi}{6}(t-2)\right) + 4$:

1. Find the amplitude, midline, period, and range.
2. When does the first maximum after $t = 0$ occur?
3. Sketch one cycle. Label a maximum, a minimum, and two points where the graph crosses its midline.

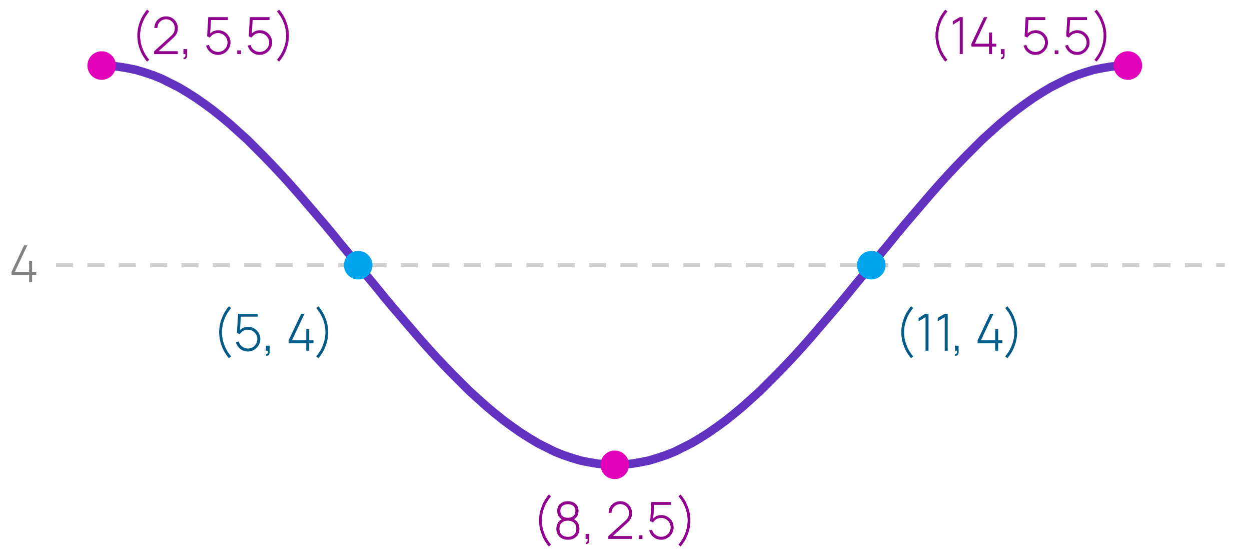
Share: explain why the period is 12, not $\pi/6$.

Read the parameters, then walk one cycle in quarter steps

METHOD

1. Match $A\cos\big(B(t-C)\big)+D$: here $A=1.5$, $B=\frac{\pi}{6}$, $C=2$, $D=4$.
2. Amplitude $|A|=1.5$, midline $y=4$, period $\frac{2\pi}{B}=12$, range $[2.5, 5.5]$.
3. Cosine peaks when its input is 0, so $t-2=0$ gives $t=2$, then every 12.
4. Step a quarter period (3) at a time to land the crossings and the minimum.

THE TRAP The period is 12, not $\frac{\pi}{6}$: B sits *inside* the cosine, so divide 2π by it.



EXIT TICKET

Work independently

1 For $p(x) = x^3 - 4x$, describe what happens on the far left and far right, and find all real roots.

2 Solve $5^{2x} = 125$.

3 For $y = 3\cos\left(\frac{\pi}{4}(t+2)\right) - 1$, state the amplitude, period, horizontal shift, and range.

RULE Write one sentence of justification for each answer. A correct number without its reason is incomplete.

WRAP-UP

What today was about

- → The **leading term** tells us what a polynomial graph does at the ends; roots are where it meets the x-axis.
- → An exponential has a **constant multiplier**. A logarithm records the exponent, and its input must be positive.
- → For large inputs: **logarithmic** < **polynomial** < **exponential** growth.
- → A trigonometric model describes repetition through **amplitude, period, shift, and midline**.

Problem Set 4

Growth comparisons, legal logarithm manipulation, and one periodic-model interpretation with written justification.

Next session: **theorem proving** — direct proof, contrapositive, contradiction, and counterexample.